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CNN-BI-LSTM-CYP: A deep learning approach for sugarcane yield prediction

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Highlights

- A Novel hybrid CNN-Bi-LSTM_CYP deep learning model to promote sustainable farming.
- Investigated the limitations of linear models to capture only linear dependencies.
- Deployed Conv-1D and Bi-LSTM layers to deal with non-linearity.
- Proposed approach outperformed the existing methods with a RMSE value of 4.05.
- Highest Sugarcane Yield expectancy is observed in the 2029year.

Article preview

Abstract

Sugarcane (Saccharum officinarum L.) is one of the principal origins of sugar and is also known as the main cash crop of India. About 19.07% of the total production of the world's sugar requirement is fulfilled by India. Traditionally, Statistical approaches have been utilized for Crop yield prediction, which is tedious and time-consuming. In this direction, the present work proposed a novel hybrid CNN-Bi-LSTM_CYP deep learning-based approach that includes convolutional layers to extract the relevant spatial information in a sequence to Bi-LSTM layers that recognize the Phenological long-term and short-term bidirectional dependencies in the dataset to predict the

Sugarcane crop yield. The experimentation was performed and validated on the historical dataset from 1950 to 2019years of the major Sugarcane-producing states of India. The preliminary results shown that the CNN-Bi-LSTM_CYP method performed well (RMSE:4.05, MSE:16.40) in comparison to traditional Stacked-LSTM (RMSE:8.8, MSE:77.79), ARIMA (RMSE:5.9, MSE:34.80), GPR (RMSE:10.1, MSE:103.3), and Holt-winter Time-series (RMSE:9.9, MSE:99.7) techniques. The study concluded that the predicted sugar yield has a minimal relative error concerning the ground truth data for the CNN-Bi-LSTM_CYP approach proving the proposed model's efficiency.

Keywords

Forecasting; Yield; Prediction; ML-GPR; CNN-Bi-LSTM; ARIMA; GRU

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Introduction

The Food and Agriculture Organization statistics signified that around 79% of the sugar is produced by processing sugarcane crops worldwide. Approximately 40% of the sugar requirement is fulfilled by Brazil, followed by India. From 2021 to 2022, about 181.18 million metric tonnes of sugar were produced, and sugar consumption exceeded 175 million metric tonnes [1]. The Population uplift of approx. 9.7 billion by 2050 will become a significant factor behind these requirement's urge worldwide. Humans will face the Consequences of this, 'Will enough food be provided at an affordable price to complete the dietary requirements?' [2], [3]. Therefore, based on these perspectives, predicting the crop before pre-harvest has become an important research domain worldwide [4]. There is a direct link between the production of Crops and their price in the market, and if there is a decrement in production can lead to a massive price hike. To overcome this demand–supply gap, there is a continuous requirement to forecast the situation of Cash crops to enhance the country's export business to a reasonable extent [5]. Several Crop Yield estimation models were proposed in the literature based on crop-related that provided reasonable outcomes [6], [7], [8]. With the emergence of the artificial intelligence field, deep learning methods, a subset of Machine learning, have overcome the limitation of traditional approaches. Machine learning (ML) and Deep learning (DL) techniques have been extensively utilized for crop yield prediction due to their ability to recognize non-linear patterns in a vast dataset [9], [10]. In this study, we built a novel CNN and Bi-LSTM hybrid deep-learning architecture for crop yield estimation of the sugarcane crop. We performed the experimentation for the study regions of India. Sugarcane Crop is one of the major cash crops of India. In 2021, approximately seventeen lakh tonnes of sugar were exported outside the country; in 2020, this amount was 4.5 lakh tonnes [11]. The latest information for the 2021 year presented that there were 479 sugar mills exists in the country that produced 77.9 tonnes of sugar [12], [13]. Therefore, in this direction, the assessment of yield and production of this crop plays a crucial role in decision-making policies for export by the farmers [6], [14].

ARIMA linear model was popular previously for time series forecasting of crops [6], [14], [15], [16], [17]. Nowadays, Deep Learning techniques overcome the Linear model issues as they cannot capture the non-linear patterns in the dataset [18], [19], [20]. Several studies signify the performance of the ARIMA model that had provided good results for linear time series forecasting and prediction. In a study, Vishwajith et al. [6] forecasted sugarcane production in India by 2020 year by considering the Area, Yield, and productivity as variables of the crop from 1951 to the 2012 year and found that the Auto-Regressive Moving average (2,1,1) model was appropriate for sugar crop experimentation. A similar approach was presented by Mishra et al. for a study of the production of Sugar crops in India using the ARIMA method. The results were evaluated using RMSE and MAPE metrics for the dataset from 1950 to 2015 [7]. The emergence of Machine learning and Deep learning techniques with remote sensing revolutionized the field of agriculture [20], [21], [22], [23], [24]. They have the capability of identification of complex relationships that exist in the patterns of the dataset among various parameters that can influence the growth of crops [25], [26], [27], [28]. In the study, Haider et al. utilized a variant of Recurrent Neural Network, i.e., LSTM (Long Short-Term Memory), an approach of Deep Learning to forecast wheat production by improvising the pre-processing phase using a smoothing function. They compared the results of RNN with the ARIMA model and found that LSTM outperformed with an R^2 value of 0.81 [29]. A similar technique of DL was used for soil moisture forecasting based on humidity and climatic factors to forecast the soil moisture in Thailand, a DNN-LSTM method was applied over the irrigation dataset that was collected with smart sensors, and the outcomes of the proposed study produced a 0.72 percent RMSE value. However, this approach cannot consecutively check the learning rate of the Adam optimizer that increased the timings of the training and testing phase of the model [30], [31]. A similar approach of LSTM was utilized in applying the recurrent neural network method to predict wheat crop yield in the Punjab region of India. The results were evaluated as RMSE 147.12 and MAE as 60.50 [32]. Many authors addressed the capabilities of machine learning methods such as Kmeans, regression model, and Naïve Bayes analysis on weather datasets to employ Regression Techniques to carry out prediction tasks with an RMSE value of 1.06 across major districts of India [33] and the comparison among ANN and regression model was discussed [34]. In this direction, Ansarifar et al. [8] applied a linear regression model for forecasting soya bean and corn crop yield by considering genotype data. They exhibited that the proposed regression model achieved 8% lower RMSE in three experimental sites and suggested hyperparameter tuning can be resolved in future experimentation. Prior to the Deep learning technique, Linear models, and Machine learning method were highly used despite their various issues [35], [36], [50]. Similarly, the study was carried out using ML to predict crop yield in the Tamil Nadu region of India using ANN techniques and found that the Random Forest method performed well in predicting accurate results by using the area and temperature information of the region [37].

The literature shown that most of the existing research in context of crop prediction were based on statistical and linear models. The existing linear models are computationally expensive as, in every iteration the method needs to determine the differencing values that enhance the training time of the method. These models do not capture the non-linear relationship among the crop attributes, and therefore, not suitable for long-term forecasting of time series data. Another constraint is to improve the training and testing time taken by machine learning and linear models to handle the large crop datasets. Recently, LSTM technique have also gained popularity in prediction and forecasting of long-term dependencies but they suffer from the Vanishing gradient issue. In this context, our work aims to propose a novel hybrid deep learning-based CNN with Bi-LSTM network that can cope up with these gaps.

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Study area

In this article, we focussed on one of India's major cash crops, i.e., the Sugarcane Crop. The investigation includes an analysis of the sugarcane crop in India's five major contributor states (Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, and Bihar). As per statistical information for the 2021 year, Uttar Pradesh is India's main contributor state of the Sugarcane Crop (approx. 48%). The description of the sugarcane crop growing and harvest period in the study regions is shown in Table S1. …

Proposed Hybrid Approach (CNN-Bi-LSTM_CYP)

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Results and discussion

The proposed approach was implemented using Python Programming in Jupyter Notebook and WEKA tool. The hardware requirements include an i5-11300H processor, an X-64-based PC, and 16GB RAM.

The Descriptive Statistics of Sugarcane crop yield (tonnes/ha) and Production (Tonnes) from 1950 to 2019 are presented for India in Table 1. The minimum yield was recorded in 1952–53 with 29.3 (tonnes/ha), and the maximum growth pattern was observed in 2017–2018 with 80.198 (tonnes/ha). Therefore, these …

Conclusion

In this study, we investigated the various linear, machine learning, and deep learning models for crop yield prediction and built the hybrid deep learning network. The CNN-Bi-LSTM_CYP approach provides a novel mechanism to overcome the limitations of linear and regression models to analyze the statistical data for time series forecasting. The efficiency of the proposed approach was also assessed on a publicly available dataset compared to other hybrid models. We have taken advantage of both the …

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Data availability

Data will be made available on request. …

CRediT authorship contribution statement

Preeti Saini: Conceptualization, Methodology, Software, Writing – original draft. **Bharti Nagpal:** Conceptualization, Methodology, Data curation, Supervision. **Puneet Garg:** Conceptualization, Visualization, Investigation. **Sachin Kumar:** Supervision, Validation, Writing – review & editing. …

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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